Computer Architecture

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Homework – 2

1)a)addi x15, x22, -45

Answer: 0xFD3B0793.

Explanation: addi instruction is of the I format which is of the form: imm[11:0](12 bits) rs1(5 bits) funct3(3 bits) rd(5 bits) opcode (7 bits).

In, addi x15, x22, -45, the immediate -45 is represented in binary as follows:

45 = 32 + 8 + 4 + 1 = 101101

-45 = 111111 010010 + 1 = 111111 010011 = 111111 010011 (imm[11:0])

22(rs1) = 16 + 4 + 2 = 10110

funct3 = 000

15(rd) = 8 + 4 + 2 + 1 = 01111

opcode = 0010011

Therefore, we get 111111010011 10110 000 01111 0010011 = 1111 1101 0011 1011 0000 0111 1001 0011 = 0xFD3B0793.

b)and x23, x8, x9

Answer: 0x00947BB3.

Explanation: and instruction is of the R format which is of the form: funct7(7 bits) rs2(5 bits) rs1(5 bits) funct3(3 bits) rd(5 bits) opcode (7 bits).

Now, funct7 = 0000000

9(rs2) = 8 + 1 = 01001

8(rs1) = 01000

funct3 = 111

23(rd) = 16 + 4 + 2 + 1 = 10111

opcode = 0110011

Therefore, we get 0000000 01001 01000 111 10111 0110011 = 0000 0000 1001 0100 0111 1011 1011 0011 = 0x00947BB3.

c)blt x2, x11, 240

Answer: 0x0EB14863.

Explanation: blt instruction is of the B format which is of the form: imm[12|10:5] rs2(5 bits) rs1(5 bits) funct3(3 bits) imm[4:1|11] opcode(7 bits).

Now, imm = 240 = 128 + 64 + 32 + 16 = 11110000 = 0(imm[12]) 0(imm[11]) 000111(imm[10:5]) 1000(imm[4:1]) 0(imm[0]).

11(rs2) = 8 + 2 + 1 = 01011

2(rs1) = 00010

funct3 = 0x4 = 100

opcode = 1100011

Therefore, we get 0 000111 01011 00010 100 1000 0 1100011 = 0000 1110 1011 0001 0100 1000 0110 0011 = 0x0EB14863.

d)sd x19, -54(x1)

Answer: 0xFD30B523.

Explanation: sd instruction is of the S format which is of the form: imm[11:5] rs2 rs1 funct3 imm[4:0] opcode.

Now, imm = -54, 54 = 32 + 16 + 4 + 2 = 000000 110110, -54 = 1111110(imm[11:5]) 01010(imm[4:0])

19(rs2) = 16 + 2 + 1 = 10011

1(rs1) = 00001

funct3 = 0x3 = 011

opcode = 0100011

Therefore, we get 1111110 10011 00001 011 01010 0100011 = 1111 1101 0011 0000 1011 0101 0010 0011 = 0xFD30B523.

e)jal x3, -10116

Answer: 0x87DFD1EF.

Explanation: jal instruction is of the J format which is of the form: imm[20|10:1|11|19:12] rd opcode.

Now, imm(21 bits) = -10116, 10116 = 8192 + 1024 + 512 + 256 + 128 + 4 = 00000010011110000100, -10116 = 11111101100001111100 = 1(imm[20]) 11111101(imm[19:12]) 1(imm[11]) 0000111110(imm[10:1]) 0(imm[0]).

3(rd) = 2 + 1 = 00011

opcode = 1101111

Therefore, we get 1 0000111110 1 11111101 00011 1101111 = 1000 0111 1101 1111 1101 0001 1110 1111 = 0x87DFD1EF.

2)a)li x5, 0xFFFFFFFFFFFFFFFF

Answer: addi x5, x0, -1

Explanation: li is a pseudo instruction that can be represented as a combination of lui + addi. Since, the immediate value here is not within the range of the addi instruction immediate value range, which is from -2048 to +2047, we cannot directly convert it using 1 addi instruction. However, we notice that 0xFFFFFFFFFFFFFFFF = -1 in unsigned form and hence we use addi x5, x0, -1 to add the value to the register x5.

b)li x5, 132

Answer: addi x5, x0, 132

Explanation: 132 is within the range -2048 to +2047, we can directly convert it using 1 addi instruction.

c)li x5, 2134

Answer:

lui x5, 0x1

addi x5, x5, 86

Explanation: 2134 is not within the range -2048 to +2047, and hence we cannot directly convert it using 1 addi instruction, and so we cannot directly convert it using 1 addi instruction. Hence, we use a lui instruction to split 2134 = 2048 + 86, apply lui for the 2048 part and regular addi for the 86 part.

d)li x5, 0x000000002345abcd

Answer:

lui x5, 0x2345b

addi x5, x5, -1075

Explanation: 2345abcd is not within the range -2048 to +2047, and hence we cannot directly convert it using 1 addi instruction. Notice, 0x2345abcd = 0x2345a000 + 0xbcd, lui for the 0x2345a000 part and addi for the 0xbcd = (16^2*11 + 16*12 + 13 = 3021 > 2047) 3021_{10} part. This will not work as the immediate isn't within the given range. We instead write it as 0x2345abcd = 0x2345b000 – 0x433 (4*16^2 + 3*16 + 3 = 1075) which can be done using an lui and an addi with the immediate value as -1075.

3)a)0x0019F233

Answer: and x4, x19, x1.

Explanation: 0x0019F233 = 0000 0000 0001 1001 1111 0010 0 0110011. Hence, the opcode for the given instruction is 0110011, which matches with all the given R format instructions. Following the R format of instructions: funct7(7 bits) rs2(5 bits) rs1(5 bits) funct3(3 bits) rd(5 bits) opcode(7 bits), we get 0000000 00001 10011 111 00100 0110011. From the values of funct7 = 0000000(0x0) and funct3 = 111(0x7), we deduce that the instruction is **and** with rd = 4, rs1 = 19, rs2 = 1. Therefore, the instruction is and x4, x19, x1.

b)0x06B4D763

Answer: bge x9, x11, 110.

Explanation: 0x06B4D763 = 0000 0110 1011 0100 1101 0111 0 1100011. Hence, the opcode for the given instruction is 1100011, which matches with all the given B format instructions. Following the B format of instructions: imm[12|10:5](1 + 6 = 7 bits) rs2(5 bits) rs1(5 bits) funct3(3 bits) imm[4:1|11](4 + 1 = 5 bits) opcode(7 bits), we get 0 000011 01011 01001 101 0111 0 1100011. From the value of funct3 = 101(0x5), we deduce that the instruction is **bge** with rs1 = 9, rs2 = 11.

The immediate value is 000000110111(deduced from the instruction) 0(The LSB of the immediate value in any branch instruction is 0 as the branch instruction can only jump to valid instructions which are surely a multiple of 2, making the LSB = 0) = 0 0000 0110 1110 = 0x06E = 6*16 + 14 = 110. Therefore, the instruction is **bge x9, x11, 110**.

c)0x0169CF93

Answer: xori x31, x19, 22.

Explanation: 0x0169CF93 = 0000 0001 0110 1001 1100 1111 1 0010011. Hence, the opcode for the given instruction is 0010011, which matches with all the given I format instructions. Following the I format of instructions: imm[11:0](12 bits) rs1(5 bits) funct3(3 bits) rd(5 bits) opcode(7 bits), we get 000000010110 10011 100 11111 0010011. From the value of funct3 = 100(0x4), we deduce that the instruction is **xori** with rd = 31, rs1 = 19. The immediate value = 0000 0001 0110 = 0x016 = 1*16 + 6 = 22. Therefore, the instruction is **xori x31, x19, 22**.